# PART III : MATHEMATICS

## SECTION – I (Total Marks : 24)

### (Single Correct Answer Type)

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

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41. Let $P(6,3)$ be a point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If the normal at the point $P$ intersects the $x-$axis at $(9, 0)$, then the eccentricity of the hyperbola is

(A) $\sqrt{\frac{5}{2}}$ (B) $\sqrt{\frac{3}{2}}$ (C) $\sqrt{2}$ (D) $\sqrt{3}$

**ANSWER : B**

42. A value of $b$ for which the equations

$$x^2+bx-1=0$$
$$x^2+x+b=0,$$

have one root in common is

(A) $-\sqrt{2}$ (B) $-i\sqrt{3}$ (C) $i\sqrt{5}$ (D) $\sqrt{2}$

**ANSWER : B**

43. Let $\omega \neq 1$ be a cube root of unity and $S$ be the set of all non-singular matrices of the form

$$\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix},$$

where each of $a$, $b$, and $c$ is either $\omega$ or $\omega^2$. Then the number of distinct matrices in the set $S$ is

(A) 2 (B) 6 (C) 4 (D) 8

**ANSWER : A**

44. The circle passing through the point $(-1, 0)$ and touching the $y-$axis at $(0, 2)$ also passes through the point

(A) $\left(-\frac{3}{2}, 0\right)$ (B) $\left(-\frac{5}{2}, 2\right)$ (C) $\left(-\frac{3}{2}, \frac{5}{2}\right)$ (D) $(-4, 0)$

**ANSWER : D**

45. If

$$\lim_{x\to 0}\left[1+x\ln(1+b^2)\right]^{\frac{1}{x}}=2b\sin^2\theta,\ b>0 \text{ and } \theta\in(-\pi,\pi],$$

then the value of $\theta$ is

(A) $\pm\frac{\pi}{4}$ (B) $\pm\frac{\pi}{3}$ (C) $\pm\frac{\pi}{6}$ (D) $\pm\frac{\pi}{2}$

**ANSWER : D**

46. Let $f:[-1,2]\to[0,\infty)$ be a continuous function such that $f(x)=f(1-x)$ for all $x\in[-1,2]$. Let $R_1=\int_{-1}^{2}x\,f(x)\,dx$, and $R_2$ be the area of the region bounded by $y=f(x)$, $x=-1$, $x=2$, and the $x$-axis. Then

(A) $R_1=2R_2$ (B) $R_1=3R_2$ (C) $2R_1=R_2$ (D) $3R_1=R_2$

**ANSWER : C**

47. Let $f(x)=x^2$ and $g(x)=\sin x$ for all $x\in\mathbb{R}$. Then the set of all $x$ satisfying $(f\circ g\circ g\circ f)(x)=(g\circ g\circ f)(x)$, where $(f\circ g)(x)=f(g(x))$, is

(A) $\pm\sqrt{n\pi}$, $n\in\{0, 1, 2,...\}$ (B) $\pm\sqrt{n\pi}$, $n\in\{1, 2,...\}$

(C) $\frac{\pi}{2}+2n\pi$, $n\in\{..., -2, -1, 0, 1, 2, ...\}$ (D) $2n\pi$, $n\in\{..., -2, -1, 0, 1, 2, ...\}$

**ANSWER : A**

48. Let $(x, y)$ be any point on the parabola $y^2=4x$. Let $P$ be the point that divides the line segment from $(0, 0)$ to $(x, y)$ in the ratio $1:3$. Then the locus of $P$ is

(A) $x^2=y$ (B) $y^2=2x$ (C) $y^2=x$ (D) $x^2=2y$

**ANSWER : C**

## SECTION – II (Total Marks : 16)

### (Multiple Correct Answer(s) Type)

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** may be correct.

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49. If

$$f(x)=\begin{cases} -x-\frac{\pi}{2}, & x\le -\frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2}<x\le 0 \\ x-1, & 0<x\le 1 \\ \ln x, & x>1, \end{cases}$$

then

(A) $f(x)$ is continuous at $x=-\frac{\pi}{2}$ (B) $f(x)$ is not differentiable at $x=0$

(C) $f(x)$ is differentiable at $x=1$ (D) $f(x)$ is differentiable at $x=-\frac{3}{2}$

**ANSWER : ABCD**

50. Let $E$ and $F$ be two independent events. The probability that exactly one of them occurs is $\frac{11}{25}$ and the probability of none of them occurring is $\frac{2}{25}$. If $P(T)$ denotes the probability of occurrence of the event $T$, then

(A) $P(E)=\frac{4}{5}$, $P(F)=\frac{3}{5}$ (B) $P(E)=\frac{1}{5}$, $P(F)=\frac{2}{5}$

(C) $P(E)=\frac{2}{5}$, $P(F)=\frac{1}{5}$ (D) $P(E)=\frac{3}{5}$, $P(F)=\frac{4}{5}$

**ANSWER : AD**

51. Let $L$ be a normal to the parabola $y^2 = 4x$. If $L$ passes through the point $(9, 6)$, then $L$ is given by

(A) $y - x + 3 = 0$ (B) $y + 3x - 33 = 0$

(C) $y + x - 15 = 0$ (D) $y - 2x + 12 = 0$

**ANSWER : ABD**

52. Let $f : (0,1) \to \mathbb{R}$ be defined by

$$f(x) = \frac{b - x}{1 - bx},$$

where $b$ is a constant such that $0 < b < 1$. Then

(A) $f$ is not invertible on $(0, 1)$

(B) $f \neq f^{-1}$ on $(0, 1)$ and $f'(b) = \frac{1}{f'(0)}$

(C) $f = f^{-1}$ on $(0, 1)$ and $f'(b) = \frac{1}{f'(0)}$

(D) $f^{-1}$ is differentiable on $(0, 1)$

**ANSWER : A**

## SECTION – III (Total Marks : 24)

### (Integer Answer Type)

This section contains **6 questions**. The answer to each of the questions is a **single-digit integer**, ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS.

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53. Let $\omega = e^{i\pi/3}$, and $a$, $b$, $c$, $x$, $y$, $z$ be non-zero complex numbers such that

$$a+b+c = x$$
$$a+b\omega+c\omega^2 = y$$
$$a+b\omega^2+c\omega = z.$$

Then the value of $\dfrac{|x|^2+|y|^2+|z|^2}{|a|^2+|b|^2+|c|^2}$ is

**ANSWER : MARKS TO ALL**

54. The number of distinct real roots of $x^4-4x^3+12x^2+x-1=0$ is

**ANSWER : 2**

55. Let $y'(x)+y(x)g'(x)=g(x)g'(x)$, $y(0)=0$, $x\in\mathbb{R}$, where $f'(x)$ denotes $\dfrac{df(x)}{dx}$ and $g(x)$ is a given non-constant differentiable function on $\mathbb{R}$ with $g(0)=g(2)=0$. Then the value of $y(2)$ is

**ANSWER : 0**

56. Let $M$ be a $3\times3$ matrix satisfying

$$M\begin{bmatrix}0\\1\\0\end{bmatrix}=\begin{bmatrix}-1\\2\\3\end{bmatrix},\quad M\begin{bmatrix}1\\-1\\0\end{bmatrix}=\begin{bmatrix}1\\1\\-1\end{bmatrix},\text{ and }M\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix}0\\0\\12\end{bmatrix}.$$

Then the sum of the diagonal entries of $M$ is

**ANSWER : 9**

57. Let $\vec{a}=-\hat{i}-\hat{k}$, $\vec{b}=-\hat{i}+\hat{j}$ and $\vec{c}=\hat{i}+2\hat{j}+3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r}\times\vec{b}=\vec{c}\times\vec{b}$ and $\vec{r}.\vec{a}=0$, then the value of $\vec{r}.\vec{b}$ is

**ANSWER : 9**

58. The straight line $2x-3y=1$ divides the circular region $x^2+y^2\le6$ into two parts. If

$$S=\left\{\left(2,\frac{3}{4}\right),\left(\frac{5}{2},\frac{3}{4}\right),\left(\frac{1}{4},-\frac{1}{4}\right),\left(\frac{1}{8},\frac{1}{4}\right)\right\},$$

then the number of point(s) in $S$ lying inside the smaller part is

**ANSWER : 2**

## SECTION – IV (Total Marks : 16)

### (Matrix-Match Type)

This section contains **2 questions**. Each question has **four statements** (A, B, C and D) given in **Column I** and **five statements** (p, q, r, s and t) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS.

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59. Match the statements given in **Column I** with the values given in **Column II**

| **Column I** | **Column II** |
|---|---|
| (A) If $\vec{a}=\hat{j}+\sqrt{3}\,\hat{k}$, $\vec{b}=-\hat{j}+\sqrt{3}\,\hat{k}$ and $\vec{c}=2\sqrt{3}\,\hat{k}$ form a triangle, then the internal angle of the triangle between $\vec{a}$ and $\vec{b}$ is | (p) $\frac{\pi}{6}$ |
| (B) If $\int_a^b (f(x)-3x)dx=a^2-b^2$, then the value of $f\left(\frac{\pi}{6}\right)$ is | (q) $\frac{2\pi}{3}$ |
| (C) The value of $\frac{\pi^2}{\ln 3}\int_{\frac{1}{6}}^{\frac{5}{6}} \sec(\pi x)\,dx$ is | (r) $\frac{\pi}{3}$ |
| (D) The maximum value of $\left\lvert Arg\left(\frac{1}{1-z}\right)\right\rvert$ for $\lvert z\rvert=1$, $z\neq 1$ is given by | (s) $\pi$ |
| | (t) $\frac{\pi}{2}$ |

**ANSWER A : q**
**B : p or p, q, r, s and t**
**C : s**
**D : t**

60. Match the statements given in **Column I** with the intervals/union of intervals given in **Column II**

| **Column I** | **Column II** |
|---|---|
| (A) The set $\left\{\text{Re}\left(\frac{2iz}{1-z^2}\right): z \text{ is a complex number, } \lvert z\rvert = 1,\ z \neq \pm 1\right\}$ is | (p) $(-\infty, -1)\cup(1, \infty)$ |
| (B) The domain of the function $f(x)=\sin^{-1}\left(\frac{8(3)^{x-2}}{1-3^{2(x-1)}}\right)$ is | (q) $(-\infty, 0)\cup(0, \infty)$ |
| (C) If $f(\theta)=\begin{vmatrix} 1 & \tan\theta & 1 \\ -\tan\theta & 1 & \tan\theta \\ -1 & -\tan\theta & 1 \end{vmatrix}$, then the set $\left\{f(\theta): 0 \le \theta < \frac{\pi}{2}\right\}$ is | (r) $[2, \infty)$ |
| (D) If $f(x)=x^{3/2}(3x-10)$, $x \ge 0$, then $f(x)$ is increasing in | (s) $(-\infty, -1]\cup[1, \infty)$ |
| | (t) $(-\infty, 0]\cup[2, \infty)$ |

**ANSWER A : s**
**B : t**
**C : r**
**D : r**